HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Built and optimised **LLM serving infrastructure** using **vLLM** and **SGLang**, improving inference throughput by **30%** for enterprise clients.
- Managed **GPU infrastructure**, including orchestration, scaling, and capacity planning for a fleet of 50+ NVIDIA A100 instances.
- Implemented **batching and quantization techniques** to optimise LLM inference performance and reduce operational costs by **18%**.
- Developed and maintained **CI/CD pipelines** using **Jenkins** and **GitLab CI** for automated, reliable model deployments to Kubernetes clusters.
- Designed and deployed **infrastructure-as-code** using **Terraform** for provisioning and managing cloud resources on AWS.
- Built comprehensive **monitoring and observability** solutions using **Prometheus** and **Grafana**, covering infrastructure health and model output degradation.
- Troubleshot production issues in **air-gapped enterprise environments**, collaborating with Applied AI and SRE teams to ensure system reliability.

SKILLS

LLMOps – vLLM, SGLang, inference engines, GPU infrastructure, batching, quantization, caching, GPU utilisation

Cloud & Infrastructure – AWS, EC2, S3, Lambda, Kubernetes, Docker, Terraform, Ansible

Observability & CI/CD – Prometheus, Grafana, Splunk, Dynatrace, Jenkins, GitLab CI, Git

Programming Languages – Python, Go, Java, SQL

Model Governance – Model registries, versioning, LLM gateways, auditing

PROJECTS

LLM Inference Platform Optimization

- Engineered a scalable inference service using **Python** and **FastAPI**, integrating **vLLM** for high-throughput LLM serving on **AWS EKS**.
- Implemented dynamic batching and model quantization strategies, achieving a **22% reduction** in p95 latency for text generation tasks across multiple models.
- Developed automated deployment scripts using **Terraform** and **Helm**, enabling self-service provisioning of inference environments for internal data science teams.

Secure Enterprise AI Deployment Framework

- Designed a framework for deploying ML models into **on-premise** and **air-gapped environments**, focusing on security and auditability.
- Built a **Python**-based model registry and versioning system that integrates with **Jenkins** for automated deployment pipelines, ensuring consistent model rollouts.
- Developed comprehensive **runbooks and documentation** for secure infrastructure management, reducing incident response time for production AI workloads by **35%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)